# OLAP Analysis with PowerBI

## Fundamentals of Business Intelligence

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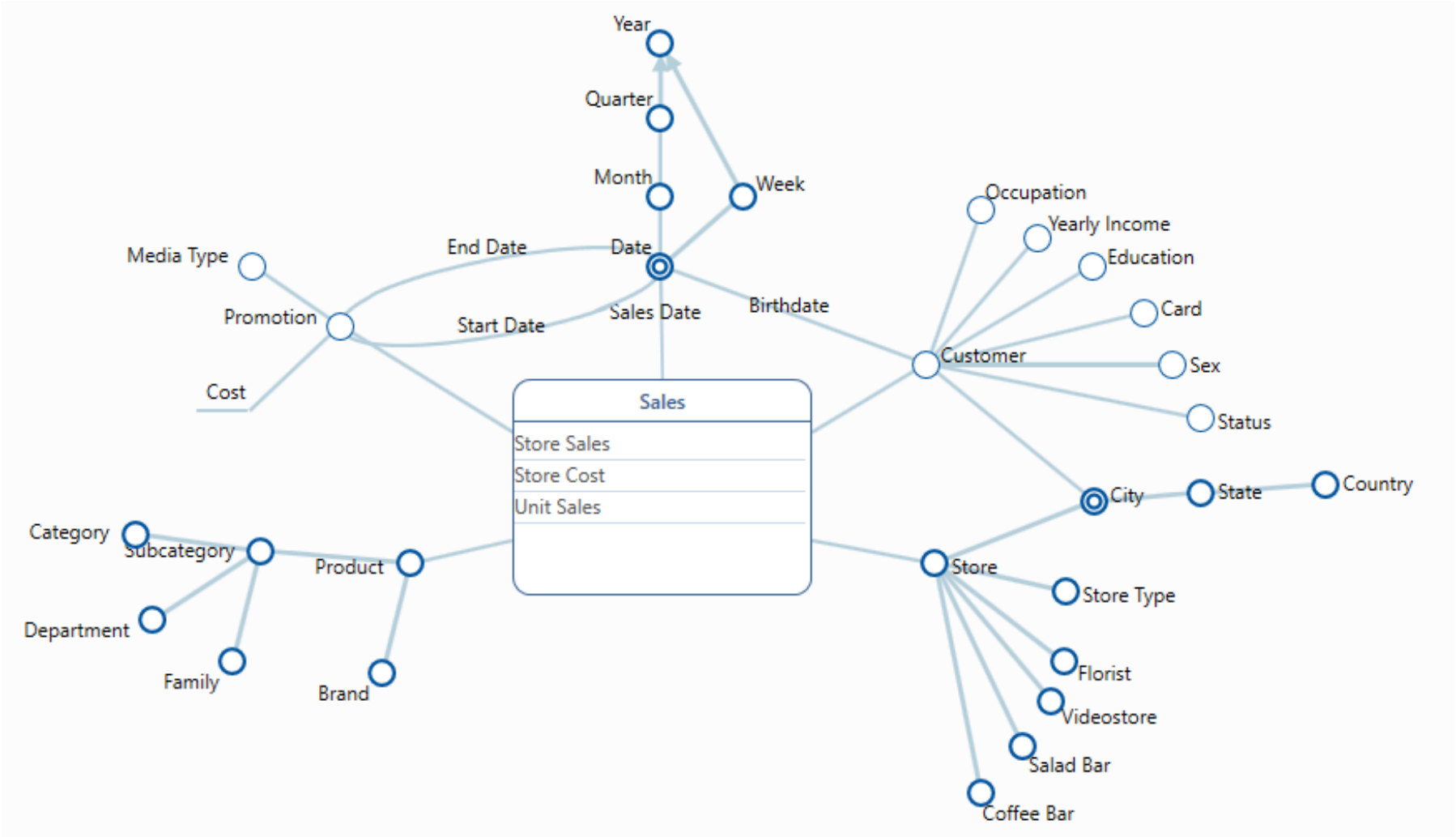
**BOLOGNA BUSINESS SCHOOL**
Alma Mater Studiorum Università di Bologna

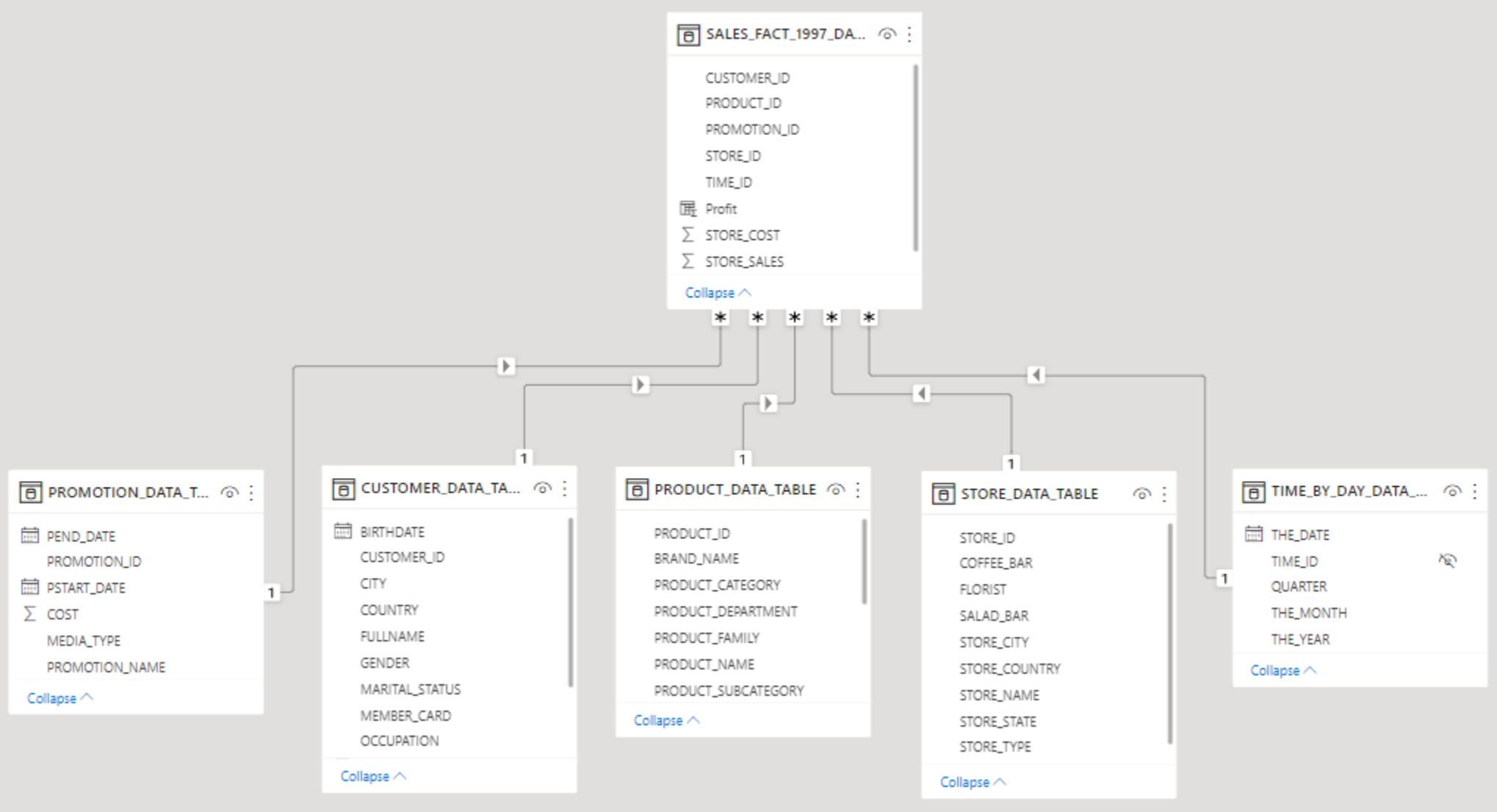
# Use case

Foodmart

# DFM – Foodmart

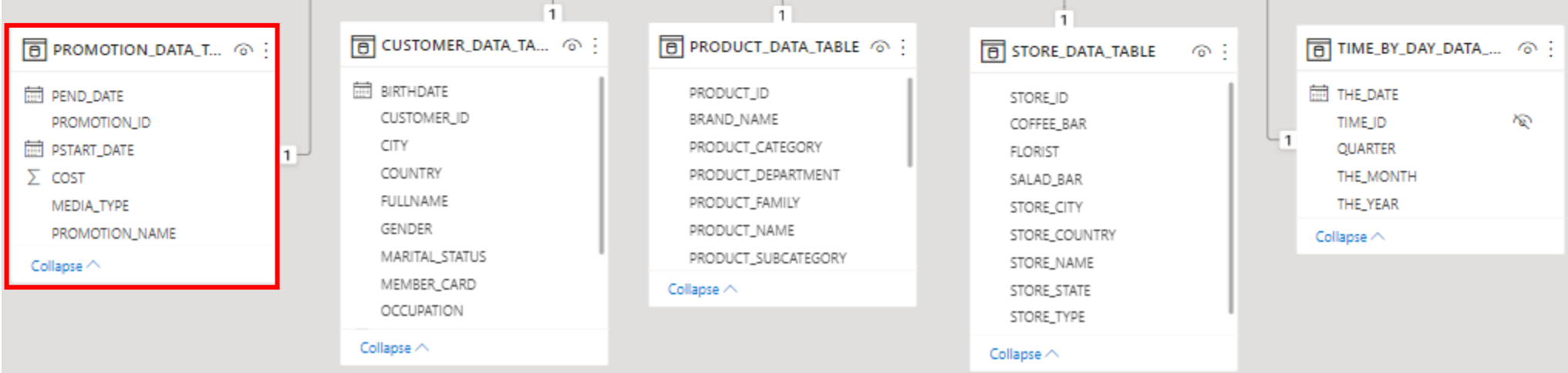


# Database Structure – Foodmart (Sales)



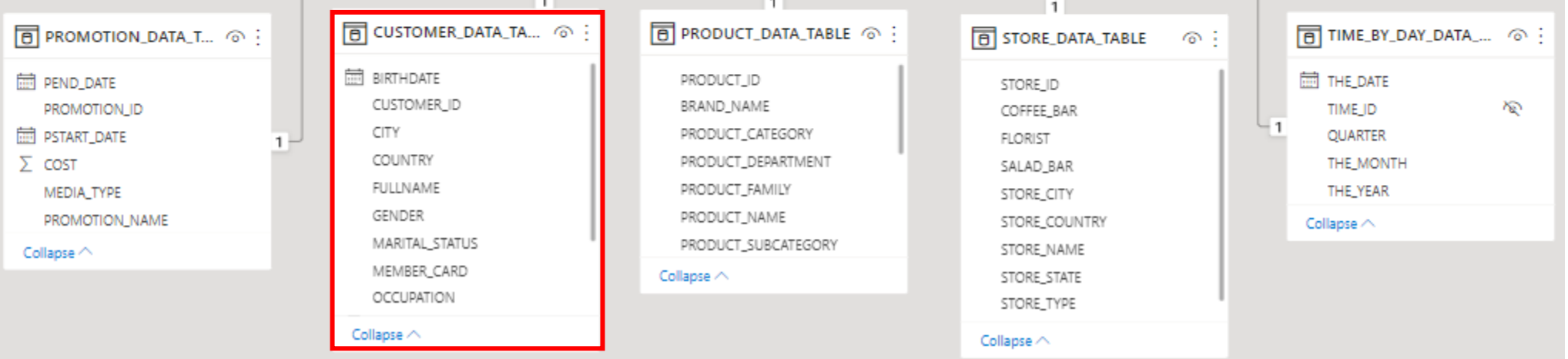
# Database Structure – Foodmart (Sales)

| | PROMOTION_ID | PROMOTION_NAME | MEDIA_TYPE | COST | PSTART_DATE | PEND_DATE |
|---|---|---|---|---|---|---|
| 1 | 532 | Weekend Markdown | Bulk Mail | 14997 | 1998-03-10 00:00:00 | 1998-03-13 00:00:00 |
| 2 | 533 | Shelf Clearing Days | Sunday Paper, Radio, TV | 10196 | 1998-03-24 00:00:00 | 1998-03-27 00:00:00 |
| 3 | 534 | Mystery Sale | Daily Paper, Radio | 6884 | 1998-04-07 00:00:00 | 1998-04-10 00:00:00 |
| 4 | 535 | Mystery Sale | Street Handout | 8527 | 1998-04-22 00:00:00 | 1998-04-23 00:00:00 |
| 5 | 536 | Bye Bye Baby | Daily Paper, Radio, TV | 5816 | 1998-05-06 00:00:00 | 1998-05-09 00:00:00 |
| 6 | 537 | Big Time Discounts | Daily Paper, Radio, TV | 14641 | 1998-05-21 00:00:00 | 1998-05-23 00:00:00 |
| 7 | 538 | Saving Days | Sunday Paper, Radio, TV | 6711 | 1998-06-04 00:00:00 | 1998-06-05 00:00:00 |
| 8 | 539 | High Roller Savings | Sunday Paper, Radio, TV | 9555 | 1998-06-17 00:00:00 | 1998-06-19 00:00:00 |
| 9 | 540 | Mystery Sale | Cash Register Handout | 8553 | 1998-07-02 00:00:00 | 1998-07-04 00:00:00 |
| 10 | 541 | Price Winners | Sunday Paper, Radio | 12251 | 1998-07-17 00:00:00 | 1998-07-18 00:00:00 |
| 11 | 542 | Green Light Special | Street Handout | 12358 | 1998-07-31 00:00:00 | 1998-08-01 00:00:00 |
| 12 | 543 | Fantastic Discounts | Radio | 13746 | 1998-08-12 00:00:00 | 1998-08-14 00:00:00 |
| 13 | 544 | Saving Days | Product Attachment | 8326 | 1998-08-28 00:00:00 | 1998-08-29 00:00:00 |
| 14 | 545 | Two for One | Daily Paper | 6897 | 1998-09-11 00:00:00 | 1998-09-14 00:00:00 |



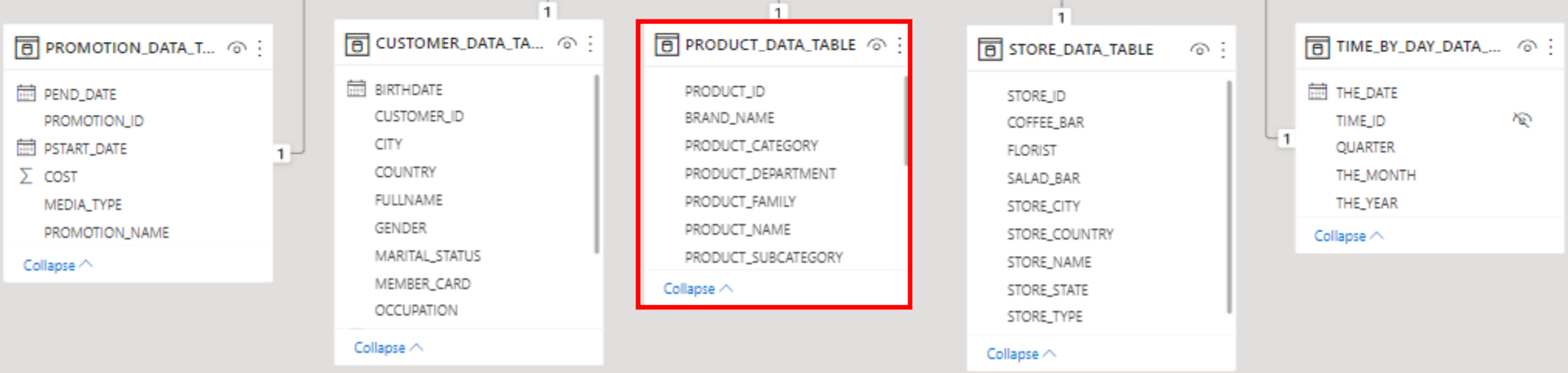
# Database Structure – Foodmart (Sales)

| | CUSTOMER_ID | CITY | STATE... | COUNTRY | BIRTHDATE | MARITAL... | YEARLY_INCOME | GENDER | MEMBER_CARD | OCCUPATION | FULLNAME | POPULATION |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 107 | Cliffside | BC | Canada | 09-MAG-21 | M | $30K - $50K | M | Bronze | Manual | Barney Velasquez | 38000000 |
| 2 | 108 | Shawnee | BC | Canada | 13-MAG-31 | M | $90K - $110K | M | Bronze | Management | Kenneth Dubois | 38000000 |
| 3 | 109 | Newton | BC | Canada | 01-AGO-29 | S | $50K - $70K | M | Bronze | Professional | Kate Maestas | 38000000 |
| 4 | 110 | West Covina | CA | USA | 03-FEB-42 | S | $10K - $30K | M | Normal | Manual | Josie Underwood | 328000000 |
| 5 | 111 | Palo Alto | CA | USA | 21-OTT-65 | M | $70K - $90K | F | Bronze | Management | Ramon Strain | 328000000 |
| 6 | 112 | Santa Monica | CA | USA | 24-NOV-48 | M | $30K - $50K | F | Bronze | Skilled Manual | Pat Azari | 328000000 |
| 7 | 113 | Sooke | BC | Canada | 20-MAG-64 | M | $110K - $130K | F | Bronze | Professional | Bob Dabit | 38000000 |
| 8 | 114 | Sooke | BC | Canada | 16-APR-43 | S | $150K + | F | Silver | Professional | Marty Carmona | 38000000 |
| 9 | 115 | Altadena | CA | USA | 09-GIU-77 | S | $90K - $110K | F | Silver | Professional | Gina Saxton | 328000000 |
| 10 | 116 | Burnaby | BC | Canada | 11-OTT-34 | S | $90K - $110K | M | Bronze | Management | Juan McLaughlin | 38000000 |
| 11 | 117 | Albany | OR | USA | 04-FEB-54 | M | $90K - $110K | F | Bronze | Professional | Carol Eyster | 328000000 |
| 12 | 118 | Bremerton | WA | USA | 04-OTT-13 | S | $50K - $70K | M | Bronze | Professional | Nancy Henry | 328000000 |
| 13 | 119 | N. Vancouver | BC | Canada | 26-LUG-57 | S | $10K - $30K | M | Normal | Manual | Chris Barros | 38000000 |
| 14 | 120 | Milwaukie | OR | USA | 28-GEN-57 | M | $30K - $50K | F | Bronze | Skilled Manual | Christopher Groome | 328000000 |
| 15 | 121 | Santa Monica | CA | USA | 17-FEB-12 | S | $30K - $50K | F | Silver | Manual | Anna Hill | 328000000 |



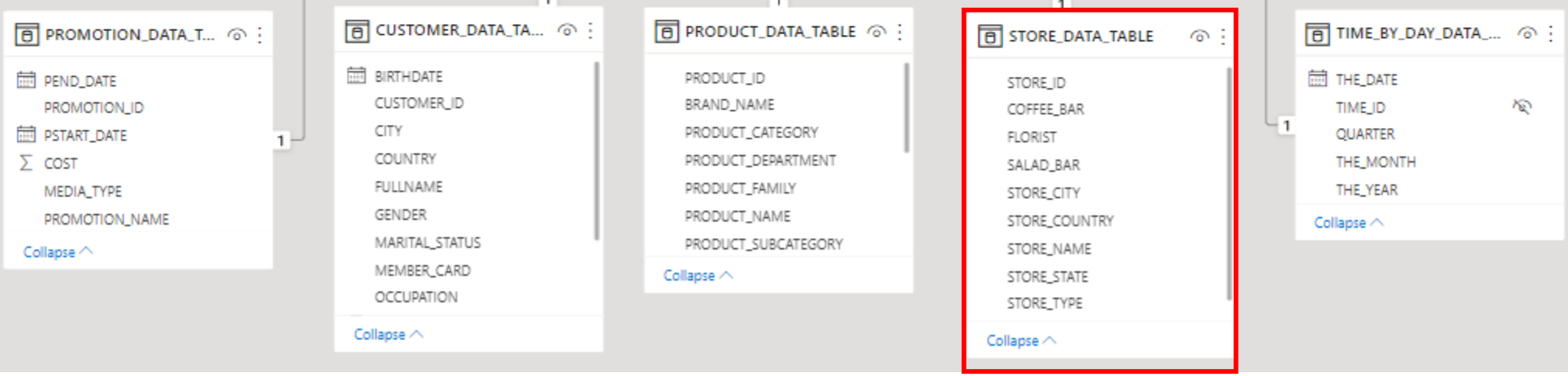
# Database Structure – Foodmart (Sales)

| | PRODUCT_ID | BRAND_NAME | PRODUCT_NAME | PRODUCT_SUBCATEGORY | PRODUCT_CATEGORY | PRODUCT_DEPARTMENT | PRODUCT_FAMILY |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Washington | Washington Berry Juice | Juice | Pure Juice Beverages | Beverages | Drink |
| 2 | 2 | Washington | Washington Mango Drink | Flavored Drinks | Drinks | Beverages | Drink |
| 3 | 3 | Washington | Washington Strawberry Drink | Flavored Drinks | Drinks | Beverages | Drink |
| 4 | 4 | Washington | Washington Cream Soda | Soda | Carbonated Beverages | Beverages | Drink |
| 5 | 5 | Washington | Washington Diet Soda | Soda | Carbonated Beverages | Beverages | Drink |
| 6 | 6 | Washington | Washington Cola | Soda | Carbonated Beverages | Beverages | Drink |
| 7 | 7 | Washington | Washington Diet Cola | Soda | Carbonated Beverages | Beverages | Drink |
| 8 | 8 | Washington | Washington Orange Juice | Juice | Pure Juice Beverages | Beverages | Drink |
| 9 | 9 | Washington | Washington Cranberry Juice | Juice | Pure Juice Beverages | Beverages | Drink |
| 10 | 10 | Washington | Washington Apple Juice | Juice | Pure Juice Beverages | Beverages | Drink |
| 11 | 11 | Washington | Washington Apple Drink | Flavored Drinks | Drinks | Beverages | Drink |
| 12 | 12 | Jeffers | Jeffers Oatmeal | Cereal | Breakfast Foods | Breakfast Foods | Food |
| 13 | 13 | Jeffers | Jeffers Corn Puffs | Cereal | Breakfast Foods | Breakfast Foods | Food |
| 14 | 14 | Jeffers | Jeffers Wheat Puffs | Cereal | Breakfast Foods | Breakfast Foods | Food |
| 15 | 15 | Jeffers | Jeffers Grits | Cereal | Breakfast Foods | Breakfast Foods | Food |

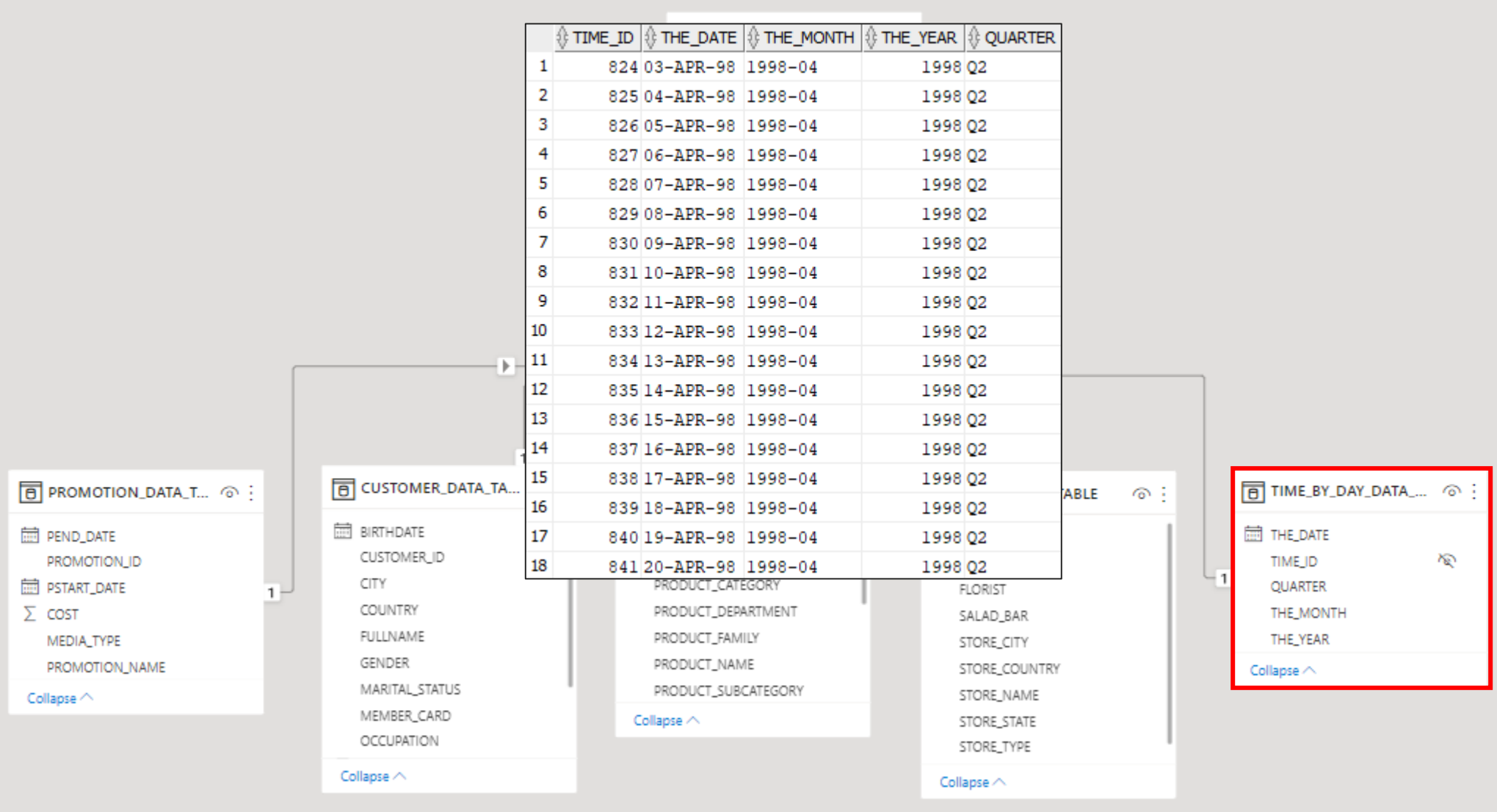


# Database Structure – Foodmart (Sales)

| | STORE_ID | STORE_TYPE | STORE_NAME | STORE_CITY | STORE_STATE | STORE_COUNTRY | COFFEE_BAR | VIDEO_STORE | SALAD_BAR | FLORIST |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | Supermarket | Store 1 | Acapulco | Guerrero | Mexico | 0 | 0 | 0 | 0 |
| 2 | 2 | Small Grocery | Store 2 | Bellingham | WA | USA | 1 | 0 | 0 | 0 |
| 3 | 3 | Supermarket | Store 3 | Bremerton | WA | USA | 0 | 0 | 1 | 0 |
| 4 | 4 | Gourmet Supermarket | Store 4 | Camacho | Zacatecas | Mexico | 1 | 0 | 1 | 1 |
| 5 | 5 | Small Grocery | Store 5 | Guadalajara | Jalisco | Mexico | 1 | 0 | 0 | 0 |
| 6 | 6 | Gourmet Supermarket | Store 6 | Beverly Hills | CA | USA | 1 | 1 | 1 | 1 |
| 7 | 7 | Supermarket | Store 7 | Los Angeles | CA | USA | 0 | 0 | 0 | 1 |
| 8 | 8 | Deluxe Supermarket | Store 8 | Merida | Yucatan | Mexico | 1 | 1 | 1 | 1 |
| 9 | 9 | Mid-Size Grocery | Store 9 | Mexico City | DF | Mexico | 0 | 0 | 0 | 0 |
| 10 | 10 | Supermarket | Store 10 | Orizaba | Veracruz | Mexico | 0 | 0 | 1 | 0 |
| 11 | 11 | Supermarket | Store 11 | Portland | OR | USA | 0 | 0 | 0 | 0 |
| 12 | 12 | Deluxe Supermarket | Store 12 | Hidalgo | Zacatecas | Mexico | 1 | 1 | 1 | 1 |
| 13 | 13 | Deluxe Supermarket | Store 13 | Salem | OR | USA | 1 | 1 | 1 | 1 |
| 14 | 14 | Small Grocery | Store 14 | San Francisco | CA | USA | 1 | 0 | 0 | 0 |
| 15 | 15 | Supermarket | Store 15 | Seattle | WA | USA | 1 | 0 | 0 | 0 |

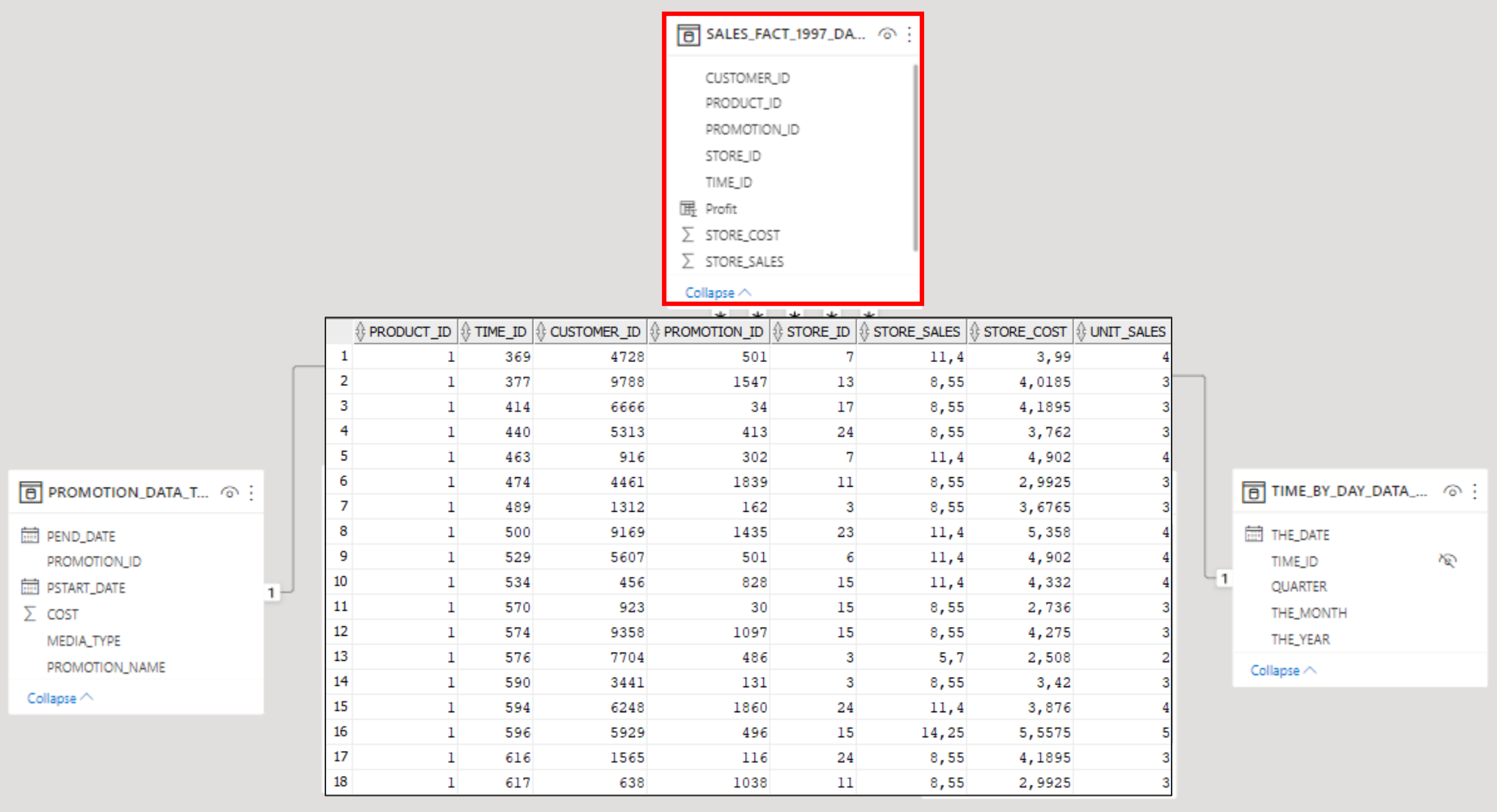


# Database Structure – Foodmart (Sales)



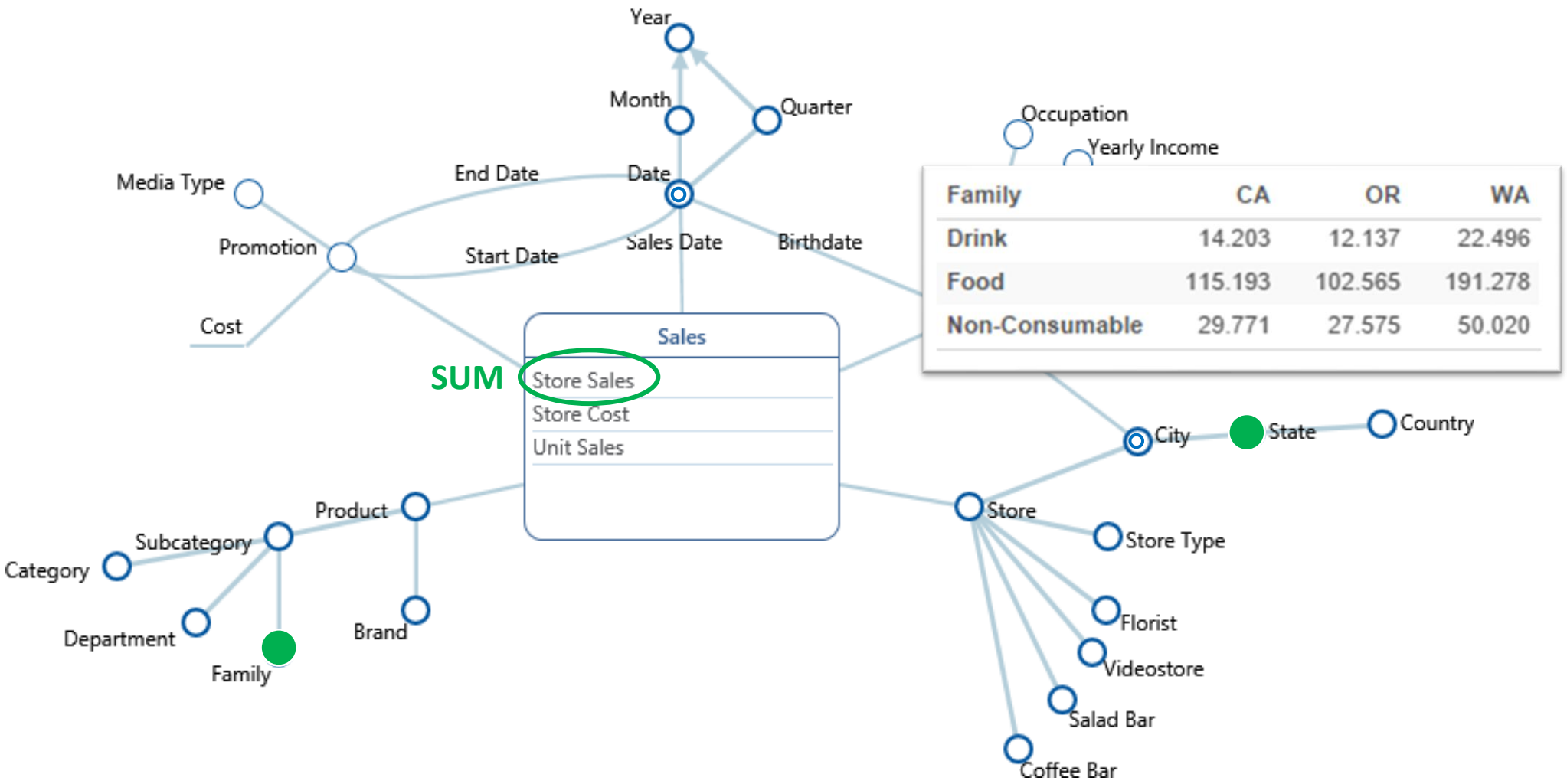

| | TIME_ID | THE_DATE | THE_MONTH | THE_YEAR | QUARTER |
|---|---|---|---|---|---|
| 1 | 824 | 03-APR-98 | 1998-04 | 1998 | Q2 |
| 2 | 825 | 04-APR-98 | 1998-04 | 1998 | Q2 |
| 3 | 826 | 05-APR-98 | 1998-04 | 1998 | Q2 |
| 4 | 827 | 06-APR-98 | 1998-04 | 1998 | Q2 |
| 5 | 828 | 07-APR-98 | 1998-04 | 1998 | Q2 |
| 6 | 829 | 08-APR-98 | 1998-04 | 1998 | Q2 |
| 7 | 830 | 09-APR-98 | 1998-04 | 1998 | Q2 |
| 8 | 831 | 10-APR-98 | 1998-04 | 1998 | Q2 |
| 9 | 832 | 11-APR-98 | 1998-04 | 1998 | Q2 |
| 10 | 833 | 12-APR-98 | 1998-04 | 1998 | Q2 |
| 11 | 834 | 13-APR-98 | 1998-04 | 1998 | Q2 |
| 12 | 835 | 14-APR-98 | 1998-04 | 1998 | Q2 |
| 13 | 836 | 15-APR-98 | 1998-04 | 1998 | Q2 |
| 14 | 837 | 16-APR-98 | 1998-04 | 1998 | Q2 |
| 15 | 838 | 17-APR-98 | 1998-04 | 1998 | Q2 |
| 16 | 839 | 18-APR-98 | 1998-04 | 1998 | Q2 |
| 17 | 840 | 19-APR-98 | 1998-04 | 1998 | Q2 |
| 18 | 841 | 20-APR-98 | 1998-04 | 1998 | Q2 |

# Database Structure – Foodmart (Sales)



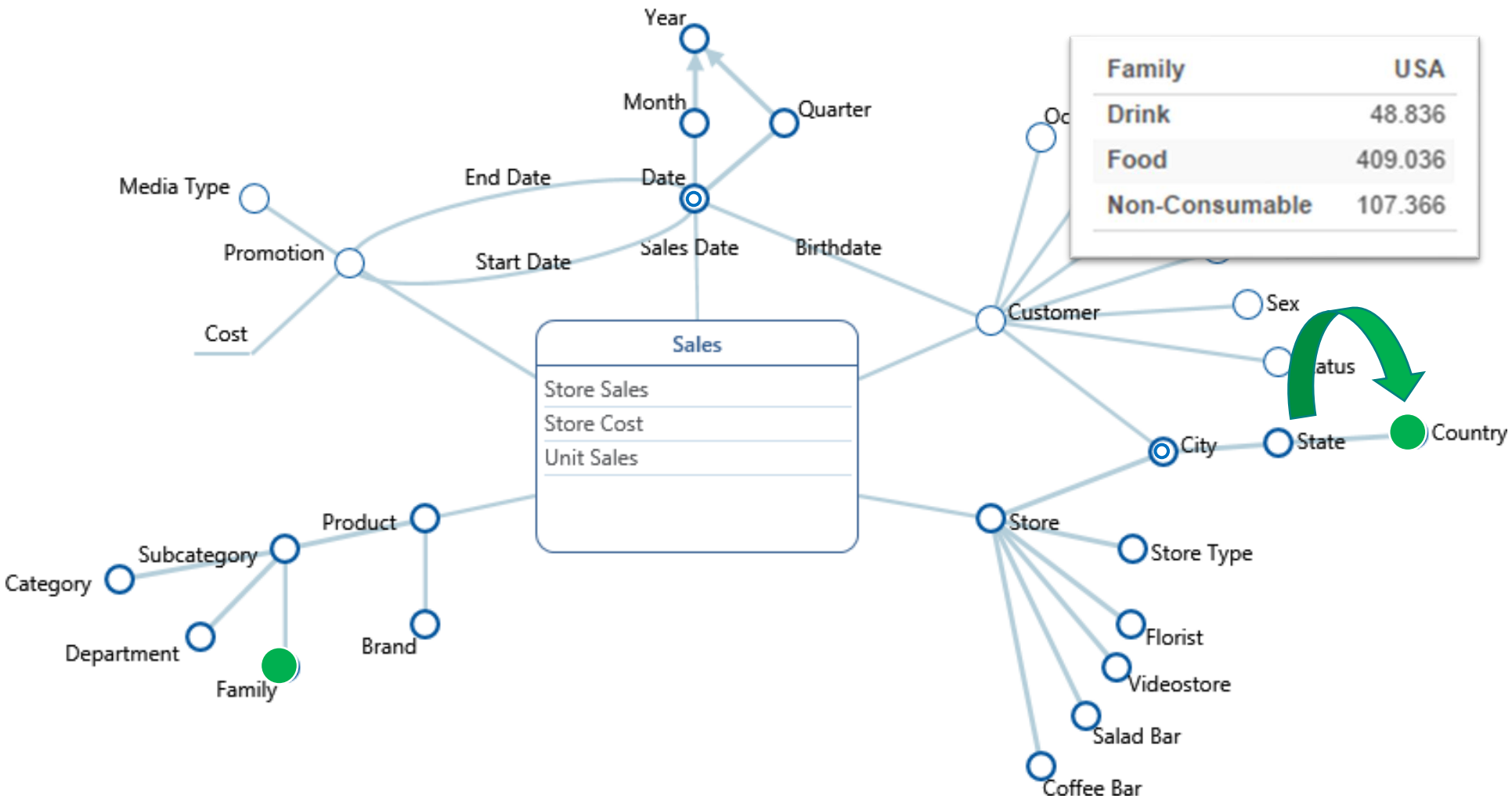

| | PRODUCT_ID | TIME_ID | CUSTOMER_ID | PROMOTION_ID | STORE_ID | STORE_SALES | STORE_COST | UNIT_SALES |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 369 | 4728 | 501 | 7 | 11,4 | 3,99 | 4 |
| 2 | 1 | 377 | 9788 | 1547 | 13 | 8,55 | 4,0185 | 3 |
| 3 | 1 | 414 | 6666 | 34 | 17 | 8,55 | 4,1895 | 3 |
| 4 | 1 | 440 | 5313 | 413 | 24 | 8,55 | 3,762 | 3 |
| 5 | 1 | 463 | 916 | 302 | 7 | 11,4 | 4,902 | 4 |
| 6 | 1 | 474 | 4461 | 1839 | 11 | 8,55 | 2,9925 | 3 |
| 7 | 1 | 489 | 1312 | 162 | 3 | 8,55 | 3,6765 | 3 |
| 8 | 1 | 500 | 9169 | 1435 | 23 | 11,4 | 5,358 | 4 |
| 9 | 1 | 529 | 5607 | 501 | 6 | 11,4 | 4,902 | 4 |
| 10 | 1 | 534 | 456 | 828 | 15 | 11,4 | 4,332 | 4 |
| 11 | 1 | 570 | 923 | 30 | 15 | 8,55 | 2,736 | 3 |
| 12 | 1 | 574 | 9358 | 1097 | 15 | 8,55 | 4,275 | 3 |
| 13 | 1 | 576 | 7704 | 486 | 3 | 5,7 | 2,508 | 2 |
| 14 | 1 | 590 | 3441 | 131 | 3 | 8,55 | 3,42 | 3 |
| 15 | 1 | 594 | 6248 | 1860 | 24 | 11,4 | 3,876 | 4 |
| 16 | 1 | 596 | 5929 | 496 | 15 | 14,25 | 5,5575 | 5 |
| 17 | 1 | 616 | 1565 | 116 | 24 | 8,55 | 4,1895 | 3 |
| 18 | 1 | 617 | 638 | 1038 | 11 | 8,55 | 2,9925 | 3 |

# OLAP query



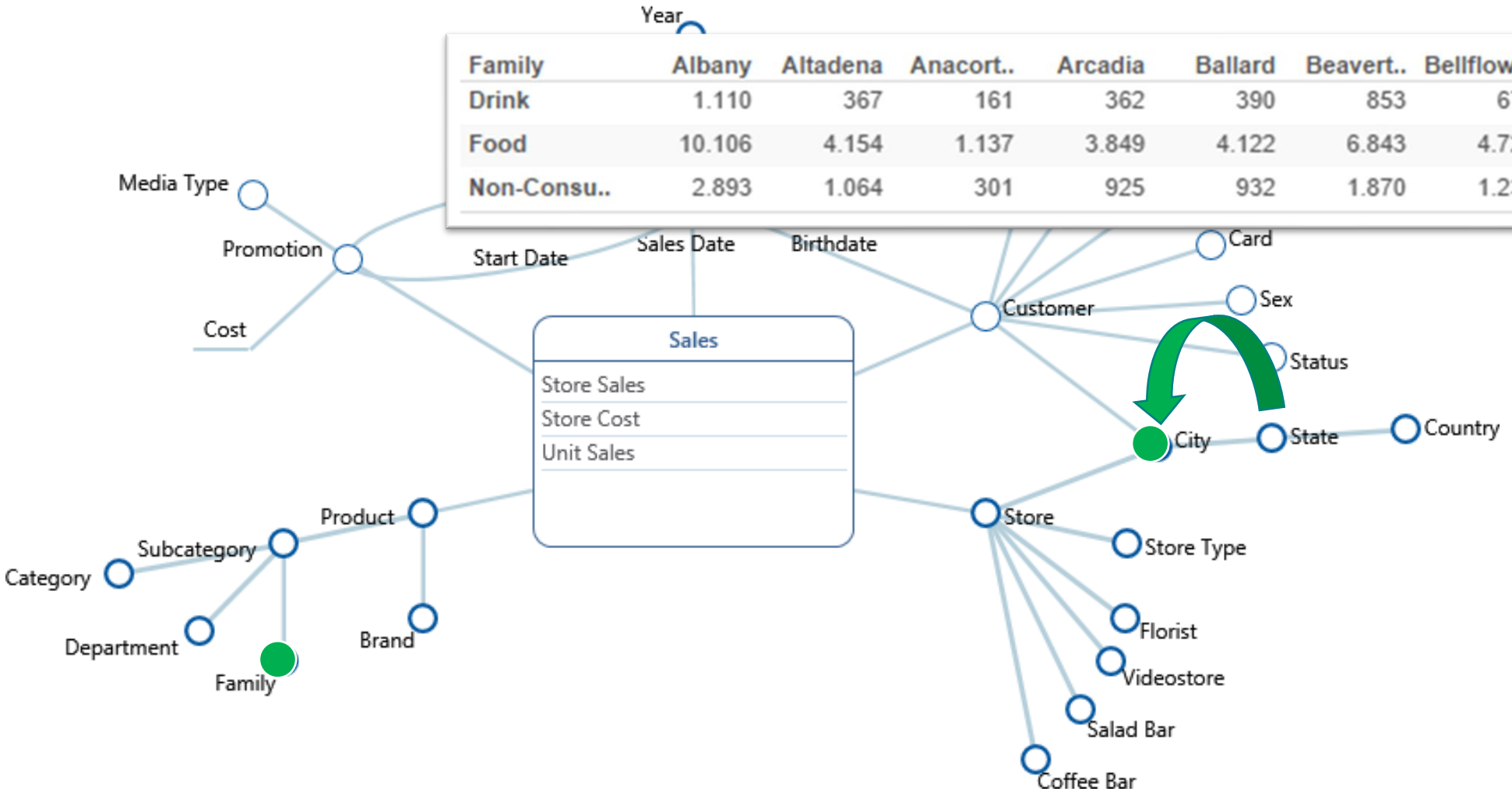

| Family | CA | OR | WA |
|---|---|---|---|
| Drink | 14.203 | 12.137 | 22.496 |
| Food | 115.193 | 102.565 | 191.278 |
| Non-Consumable | 29.771 | 27.575 | 50.020 |

# OLAP Operators: Roll-Up



# OLAP Operators: Drill-down

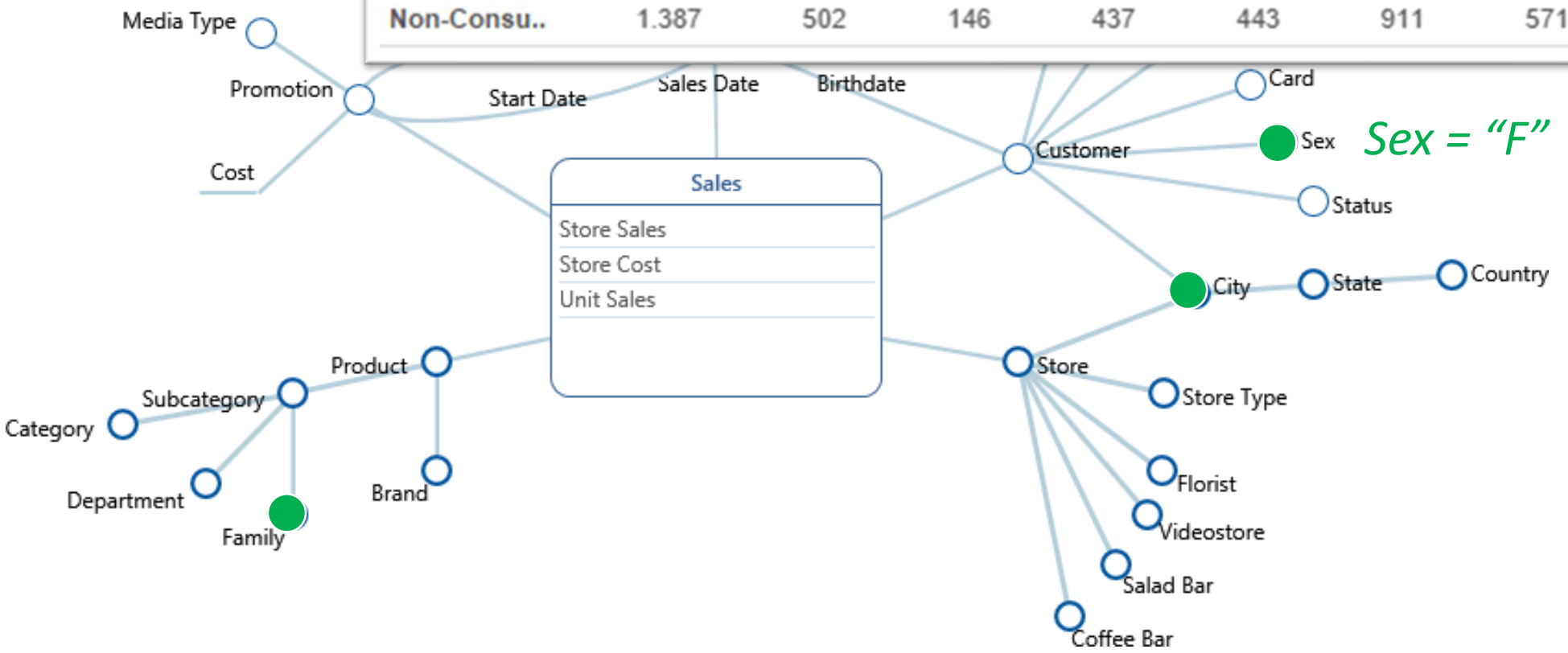
| Family | Albany | Altadena | Anacort.. | Arcadia | Ballard | Beavert.. | Bellflower |
|---|---|---|---|---|---|---|---|
| Drink | 1.110 | 367 | 161 | 362 | 390 | 853 | 673 |
| Food | 10.106 | 4.154 | 1.137 | 3.849 | 4.122 | 6.843 | 4.722 |
| Non-Consu.. | 2.893 | 1.064 | 301 | 925 | 932 | 1.870 | 1.238 |



# OLAP Operators: Slice & Dice

Year

| Family | Albany | Altadena | Anacort.. | Arcadia | Ballard | Beavert.. | Bellflower |
|---|---|---|---|---|---|---|---|
| Drink | 568 | 158 | 81 | 153 | 257 | 351 | 367 |
| Food | 4.270 | 1.883 | 555 | 1.960 | 1.999 | 3.026 | 2.477 |
| Non-Consu.. | 1.387 | 502 | 146 | 437 | 443 | 911 | 571 |



# Exercises - Part I

Setup the data on PowerBI

# Exercise 0

- Open the CSV files on PowerBI
- Setup the relationships between the files
- Clean the data
  - Check the month
  - Remove wrong sales
  - IDs or names?
  - Hide unused fields
- Create hierarchies

# Exercises - Part II

Familiarize with PowerBI querying functionalities

# Exercise 1

- Use a bar chart to plot the total sum of *STORE_SALES* for each *STORE_STATE*
  - Which one is the state with the highest sales?
- Apply a drill-down operation to show the sales at the *STORE_CITY* level
  - Are there cities whose sales are much lower than the others'?
- How many stores are there in each *STORE_STATE*? In each *STORE_CITY*?
  - Color the bars based on the *Count(Distinct)* summarization function over the *STORE_NAME* attribute
  - Would it be reasonable to say that cities with fewer stores also have lower total sales?

# Exercise 2

- Use a bar chart to plot the total sum of *STORE_SALES* by *STORE_CITY* and assign the *STORE_TYPE* to the Legend property
  - Can you notice any interesting pattern?
- Use a bar chart to plot the total sum of *STORE_SALES* by *STORE_TYPE*
  - Assign the number of stores to the color property
  - Is the result surprising/expected?

# Exercise 3

- Use a line chart to plot the monthly sales trend
  - Any interesting pattern?
- Split the previous chart by *STORE_STATE*
  - Put the STORE_STATE in the Legend
  - Does the previous pattern hold for each state?
- Visualise the impact of each *STORE_FAMILY* on the total sales while still showing the monthly trends
  - Use a Stacked area chart, where the *STORE_STATE* is in the Small multiples and the *PRODUCT_FAMILY* in the Legend

# Exercise 4

- Analyze sales by *STORE_TYPE* (sorted by descending order)
- Drill-down to the stores
- Add the number of customers
  - Use the *Count(Distinct)* summarization function
  - In case of wrong calculation (i.e., if you get the same value in all rows):
    - Go back to the Model
    - Double-click the relationships between CUSTOMER and SALES
    - Set the *Cross filter direction* to *Both*
- Add the average sales per customer
  - Create a new measure, calculated by dividing the sum of store sales by the count of distinct customers

# Exercise 5

- Create a table to visualize the sales for each *OCCUPATION* (*Customer* dimension)
- Exclude (i.e., filter out) the tuples where the value of *STORE_SALES* is lower than 5
- Apply another filter (in addition to the previous one) to exclude all occupations where the total sales is lower than 80K

# Exercise 6

- Create a table to visualize with the top ten customers by total sales
  - Show both *CUSTOMER_ID* and *FULLNAME*
- Add the *Occupation* field
- Turn it into a matrix (without the *FULLNAME*)
- Add a measure on the Customer table calculating a ranking of customers
  - First, declare a new measure simply calculating the sum of *STORE_SALES*
  - Then, declare a new measure calculating the RANKX, where
    - The 1st parameter is the attribute that we want to order, i.e., the *CUSTOMER_ID*
    - The 2nd parameter is the measure to be used for ordering, i.e., the one declared above
- Take the first ten customers for each occupation by filtering on the rank

# Exercise 7

- Create a histogram of StoreSales
  - Right-click on *STORE_SALES* > New group > Create bins of size 2
  - Create a bar chart showing the count of records for each bin
- Use the same binning to plot a bar chart with the average *STORE_COST* for each bin
  - Do you see a correlation in the data?
- Plot the same result as a scatter chart
  - Find the chart in the list of visuals
  - Put *STORE_COST* and *STORE_SALES* on X and Y axis, respectively (without summarizing)

# Exercise 8

- Create a new column calculating the profits
  - *PROFIT = STORE_SALES - STORE_COST*
- Create a line chart showing the monthly trend of profits, sales, and costs

# Part III

Open exploration of the cube

# Exercise 9

- Goal: describe sales from the perspective of customers
- Some hints:
  - Check distribution of sales (or profits) with respect to different attributes
  - Are there correlations between sales and number of customers?
  - Create bins where necessary (e.g., population)
  - Try some combinations of attributes (e.g., with the stacked bar chart)
  - Calculate the age of customers from their birthdate
    - A new column must be defined
    - <newColumn> = DATEDIFF(<date1>, <date2>, YEAR)

# Exercise 10

- Goal: correlate sales between customer and product attributes
- Some hints:
  - Are sales of the different product families equally distributed among genders?
    - Explore the whole hierarchy using drill-down and roll-up functionalities
  - Rank brands by yearly income of customers
    - Show the top-3 brands for each yearly income
    - Plot ranks in a matrix
      - Possibly add "Background color" to "Cell elements" to make it a heatmap

# References

Doc: [https://docs.microsoft.com/en-us/power-bi/fundamentals/](https://docs.microsoft.com/en-us/power-bi/fundamentals/)

A *lot* of YouTube videos